Why should you care about Bitcoin Core?

@glozow

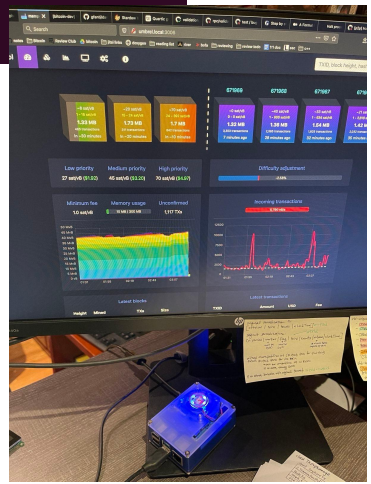
6B00 2C6E A3F9 1B1B 0DF0 C9BC 8F61 7F12 00A6 D25C

As a Bitcoin user...

- How do you see how much money you have?
- How do you know your transaction was sent?

```
[redacted]:~/bitcoin$ src/bitcoin-cli getblockchaininfo
{
  "chain": "main",
  "blocks": 818853,
  "headers": 818853,
  "bestblockhash": "00000000000000000017f784438e38b2ff8c747d8adffa10aed5e3343d4ab5f",
  "difficulty": 67957790298897.88,
  "time": 1701178265,
  "mediantime": 1701176776,
  "verificationprogress": 0.9999996079878494,
  "initialblockdownload": false,
  "chainwork": "0000000000000000000000000000000000000000005ed9b8b7a01b0a34611ca1a0",
  "size_on_disk": 10359899411,
  "pruned": true,
  "pruneheight": 813469,
  "automatic_pruning": true,
  "prune_target_size": 10485760000,
```



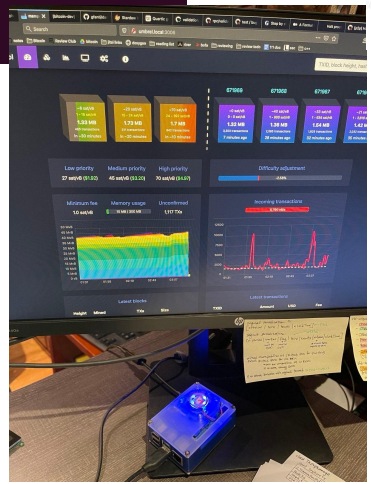


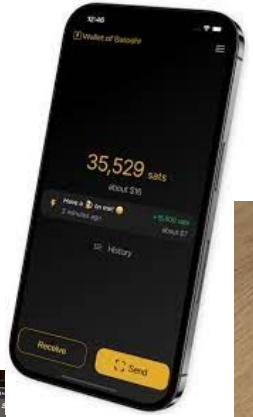
run your
own node?

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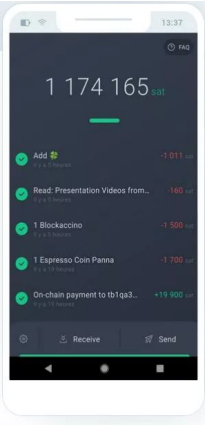


run your
own node?





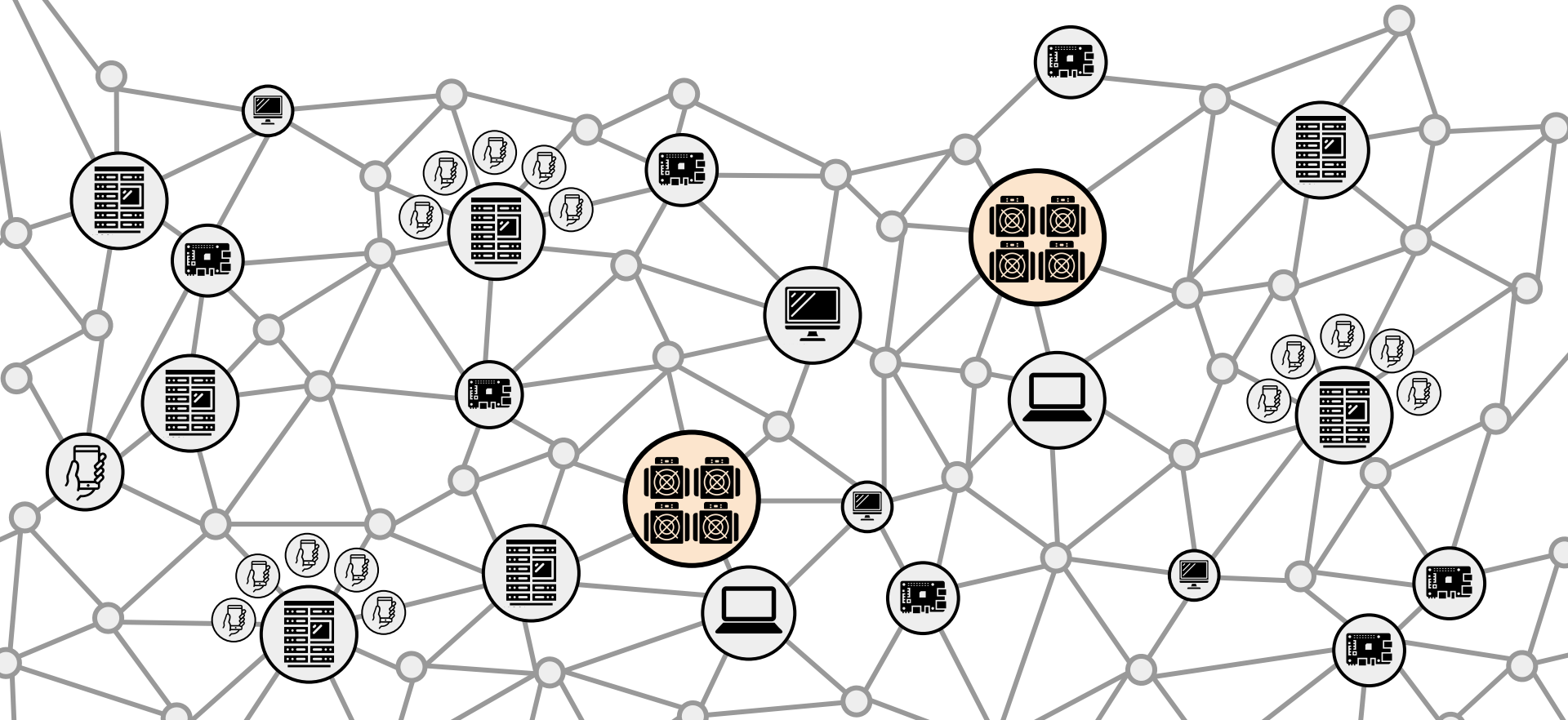




{custodial, non-custodial}
mobile wallets?

exchanges?

ultimately,
you connect to the Bitcoin p2p network:



98.5% of nodes on the network run Bitcoin Core





bitcoind might be:

- on a server supporting 10000s of users
- on a personal laptop
- on a raspberry pi

and it will:

- connect to anons who may be attackers
- handle people’s money
- be spied upon

1. reference protocol implementation

2. (probably) your node

“Isn’t the protocol done?”

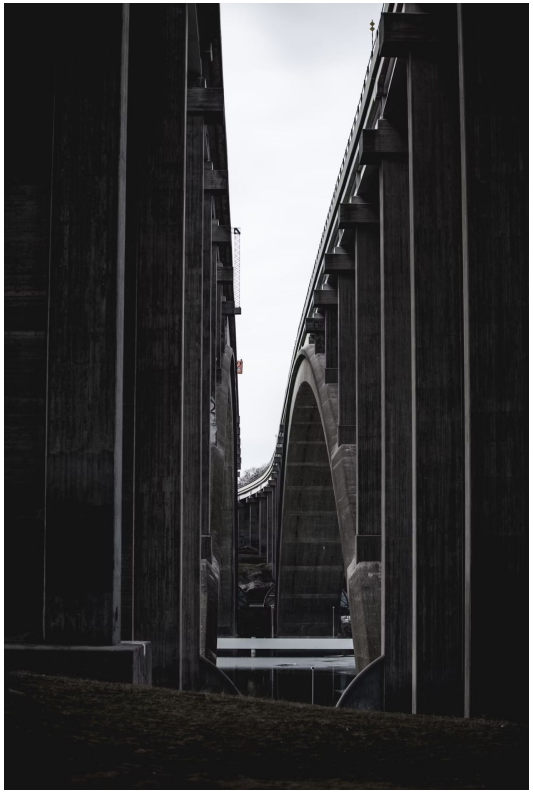



just enough for a
few nerds to cross
privacy if you're
careful / lucky
will break if 1 rat
chews on it

scalable and accessible







robust and secure against attackers

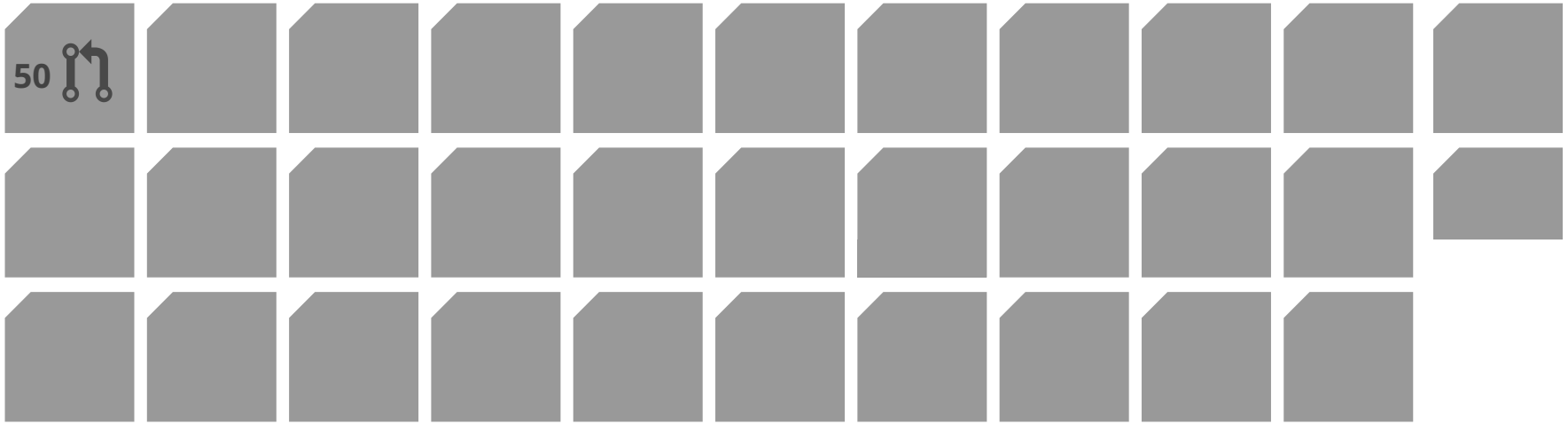


very private by default

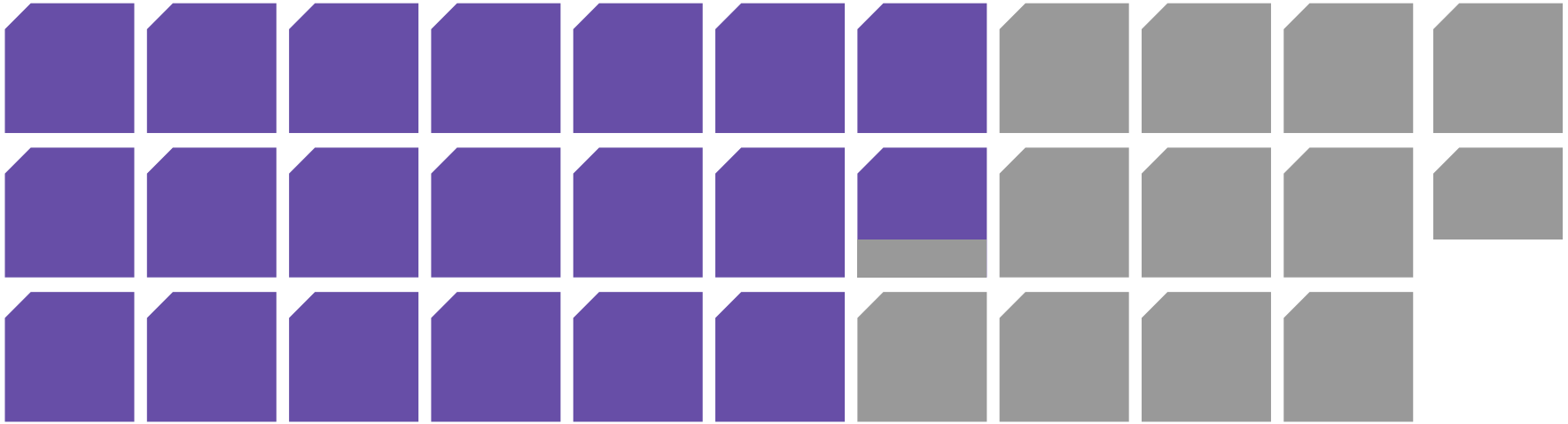


“Can the devs DO something?”

1586 pull requests opened to bitcoin in the past year...

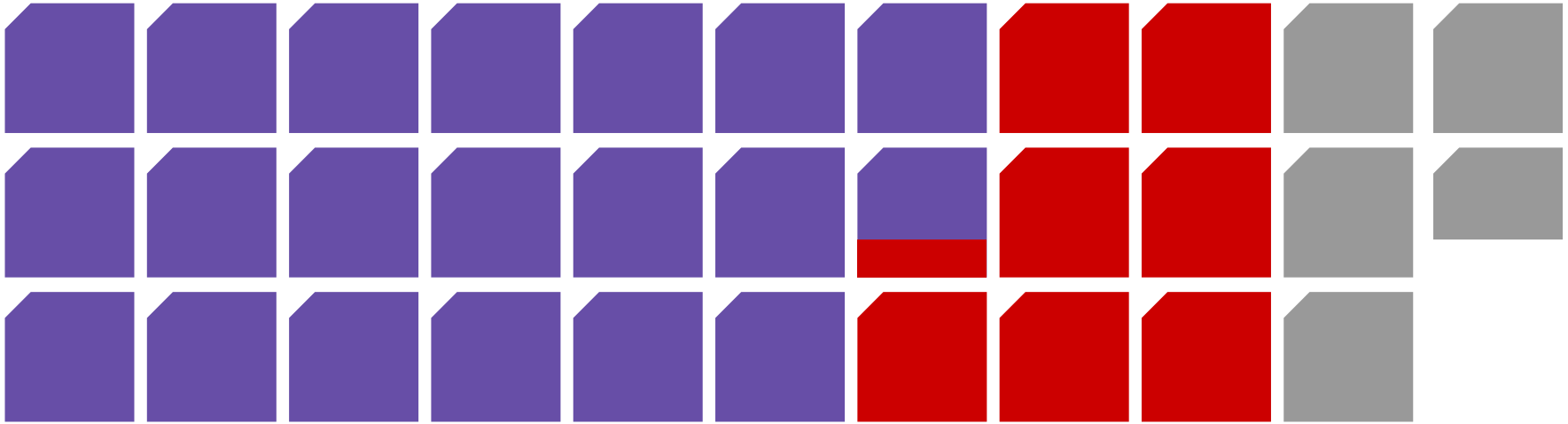


1586 pull requests opened to bitcoin in the past year...



990 merged

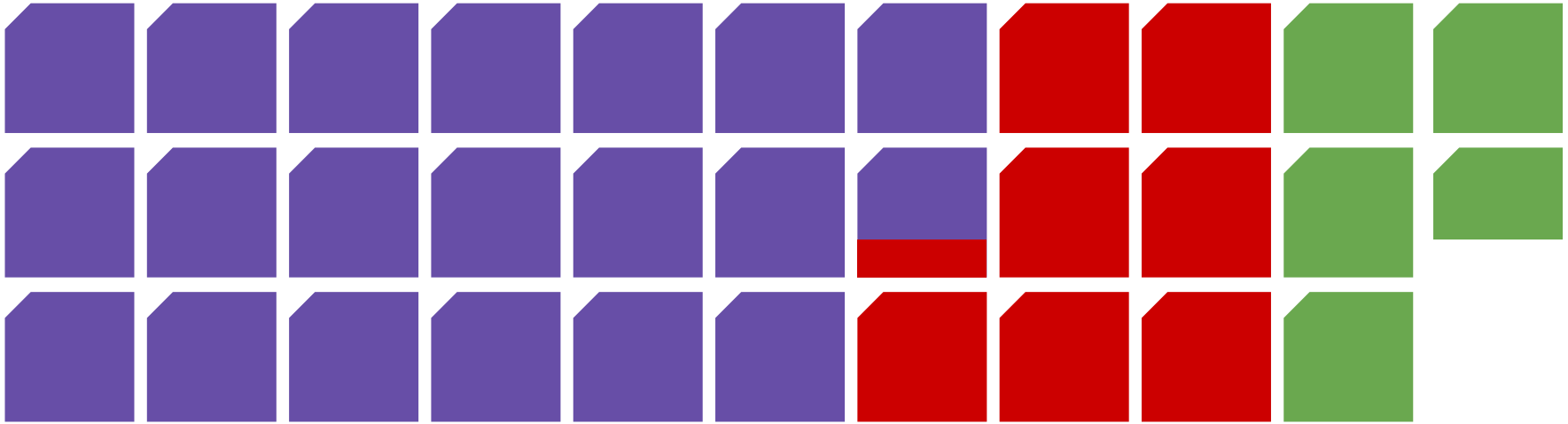
1586 pull requests opened to bitcoin in the past year...



990 merged

369 closed

1586 pull requests opened to bitcoin in the past year...



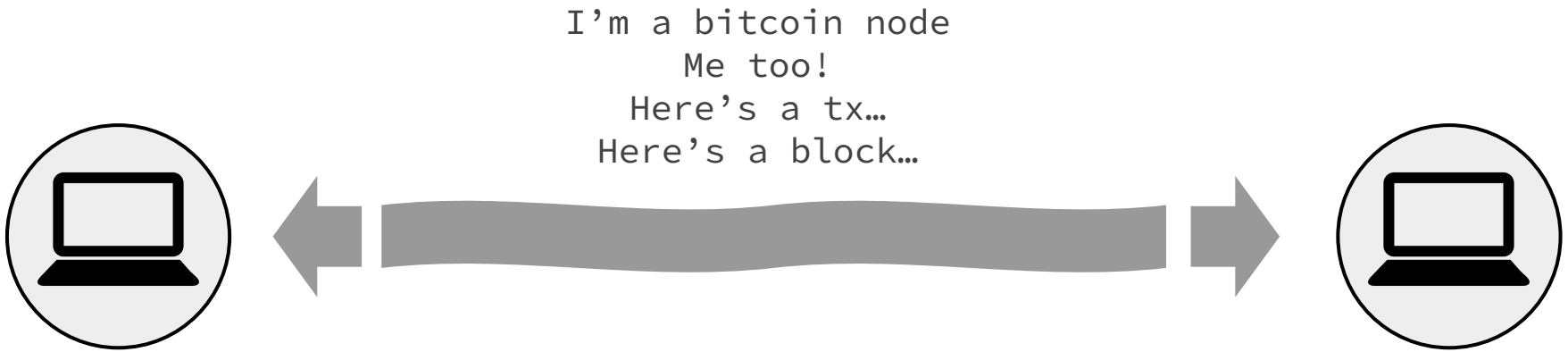
990 merged

369 closed

227 open

BIP324 encrypted p2p connections

PRs #27479, #28008, #28196, #28331, etc



BIP324 encrypted p2p connections

PRs #27479, #28008, #28196, #28331, etc

I’m a bitcoin node
Me too!
Here’s a tx…
Here’s a block…



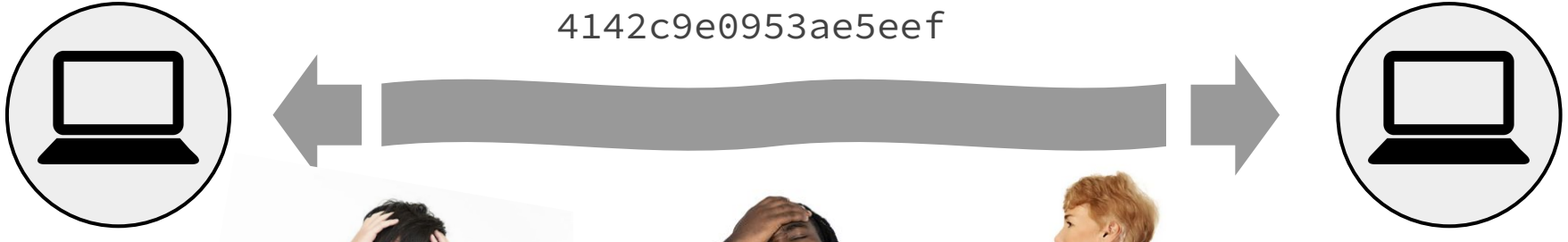




BIP324 encrypted p2p connections

PRs #27479, #28008, #28196, #28331, etc

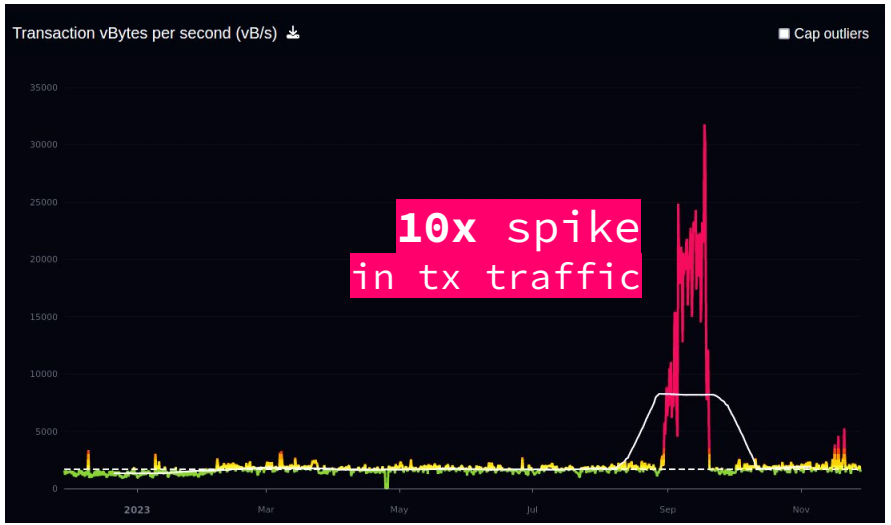
```
9e11f19e9f49b4fc6
07bcd2c51313a6efe
f81daa4fda1c39f0f
4142c9e0953ae5eef
```





More Examples

- more robust block downloads
  PR #27626
- more scalable transaction relay
  PR #27610

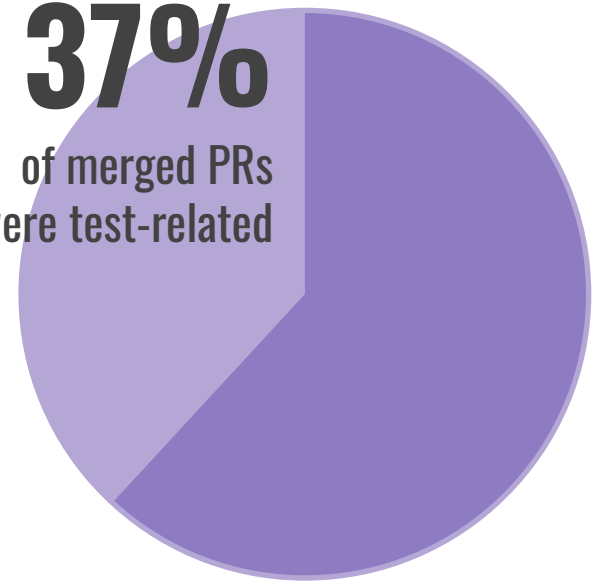


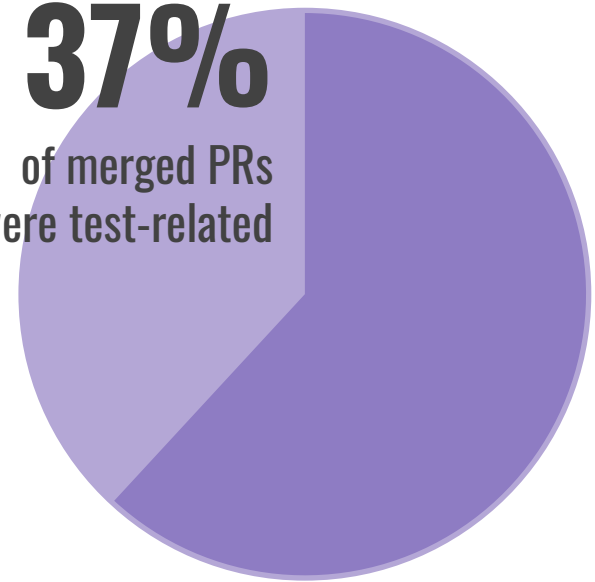
More Examples

- more robust block downloads
  PR #27626
- more scalable transaction relay
  PR #27610
- isolation of consensus engine (libbitcoinkernel)
  PRs #28113, #27125, #27636, etc.
- coverage-guided automatic testing
  PRs #28450, #26662, #27585, #26898, etc.

37%

of merged PRs
were test-related





There's still lots more to do!

Current view:	top level		Hit	Total	Coverage
Test:	total_coverage.info	Lines:	78186	94377	82.8 %
Date:	2023-11-17 15:17:47	Functions:	15301	19229	79.6 %
		Branches:	143677	295727	48.6 %

Directory	Line Coverage		Functions		Branches	
src	88.5 %	21089 / 23824	84.2 %	8184 / 9723	53.1 %	21046 / 39623
src/common	92.1 %	877 / 952	88.0 %	132 / 150	58.7 %	1095 / 1864
src/compat	85.0 %	17 / 20	100.0 %	6 / 6	75.0 %	3 / 4
src/consensus	98.1 %	204 / 208	97.7 %	42 / 43	70.5 %	186 / 264
src/crypto	71.2 %	2051 / 2882	78.6 %	187 / 238	77.2 %	281 / 364
src/index	85.3 %	678 / 795	94.6 %	106 / 112	49.6 %	446 / 899
src/init	93.3 %	83 / 89	92.9 %	13 / 14	50.6 %	159 / 314
src/interfaces	42.9 %	18 / 42	40.4 %	19 / 47	0.0 %	0 / 2
src/kernel	96.5 %	699 / 724	93.7 %	104 / 111	53.6 %	390 / 727
src/logging	100.0 %	27 / 27	100.0 %	12 / 12	61.5 %	16 / 26
src/node	84.4 %	2010 / 2382	77.2 %	308 / 399	54.1 %	1887 / 3486
src/policy	94.2 %	801 / 850	92.9 %	91 / 98	62.5 %	627 / 1004
src/primitives	93.6 %	248 / 265	96.0 %	389 / 405	55.9 %	132 / 236
src/rpc	94.3 %	6034 / 6402	97.4 %	487 / 500	48.1 %	12164 / 25290
src/script	95.3 %	5395 / 5660	86.3 %	953 / 1104	65.0 %	6693 / 10295
src/support	92.8 %	168 / 181	86.8 %	33 / 38	69.3 %	61 / 88
src/support/allocators	94.3 %	83 / 88	51.2 %	66 / 129	78.6 %	22 / 28
src/test	98.3 %	20066 / 20409	99.2 %	1794 / 1808	49.1 %	71039 / 144615
src/test/fuzz	0.0 %	0 / 8122	0.0 %	0 / 1385	0.0 %	0 / 11165
src/test/fuzz/util	0.0 %	0 / 196	0.0 %	0 / 31	0.0 %	0 / 112
src/test/util	78.1 %	683 / 875	64.2 %	113 / 176	43.3 %	529 / 1222
src/univalue/include	95.4 %	104 / 109	93.4 %	85 / 91	82.9 %	63 / 76
src/univalue/lib	95.7 %	512 / 535	100.0 %	42 / 42	75.1 %	392 / 522
src/univalue/test	100.0 %	390 / 390	100.0 %	14 / 14	50.3 %	585 / 1164
src/util	80.2 %	1441 / 1796	79.0 %	664 / 841	59.7 %	994 / 1665
src/wallet	84.3 %	7771 / 9219	83.9 %	963 / 1148	52.3 %	7735 / 14779
src/wallet/rpc	95.5 %	3879 / 4063	98.0 %	194 / 198	47.1 %	9252 / 19663
src/wallet/test	97.7 %	2582 / 2642	89.1 %	237 / 266	50.1 %	7631 / 15236
src/wallet/test/fuzz	0.0 %	0 / 324	0.0 %	0 / 35	0.0 %	0 / 514
src/zmq	90.2 %	276 / 306	96.9 %	63 / 65	51.9 %	249 / 480

Generated by: LCOV version 1.16

the good news

- devs are
  - working on privacy, scalability, security, decentralization
  - fixing vulnerabilities

the good news

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  - fixing vulnerabilities

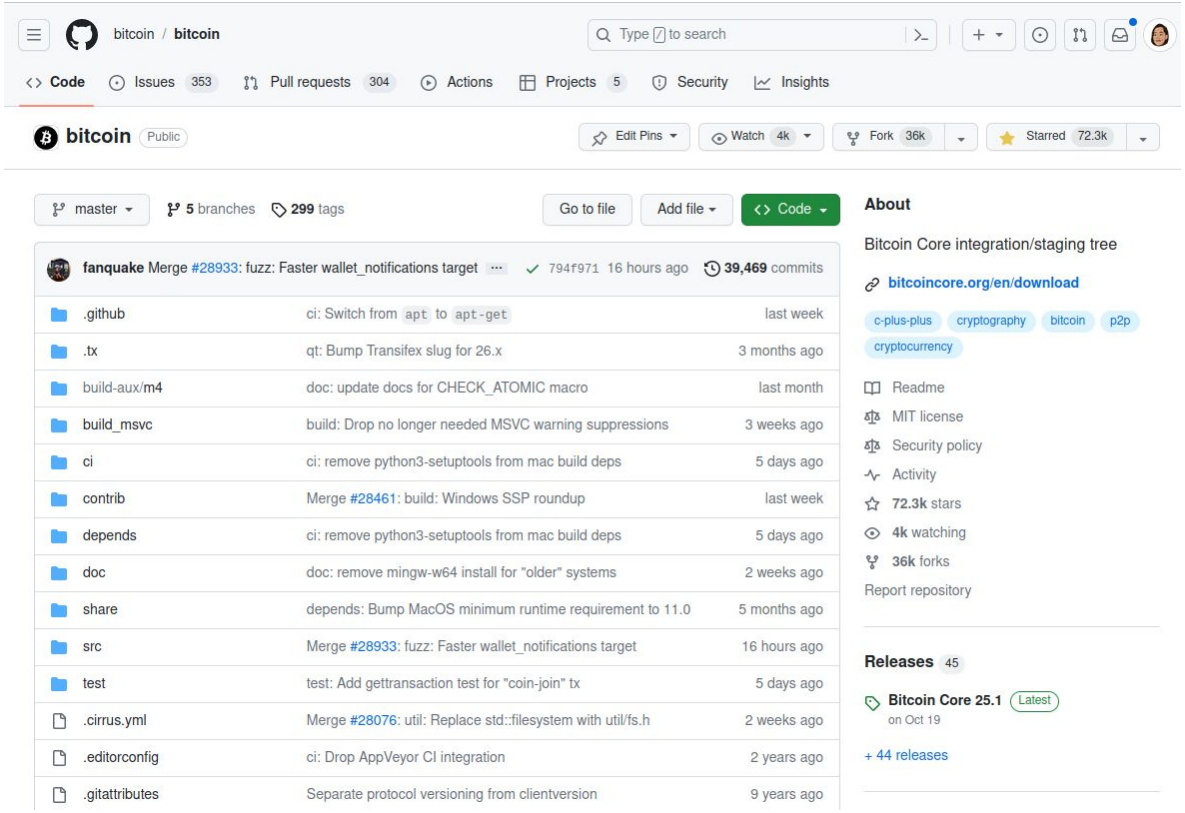
the bad news

- in the last year, 46 users committed to the repo
  - 31 with >= 10
  - 11 with >= 50

“Why aren’t they adding more devs?”

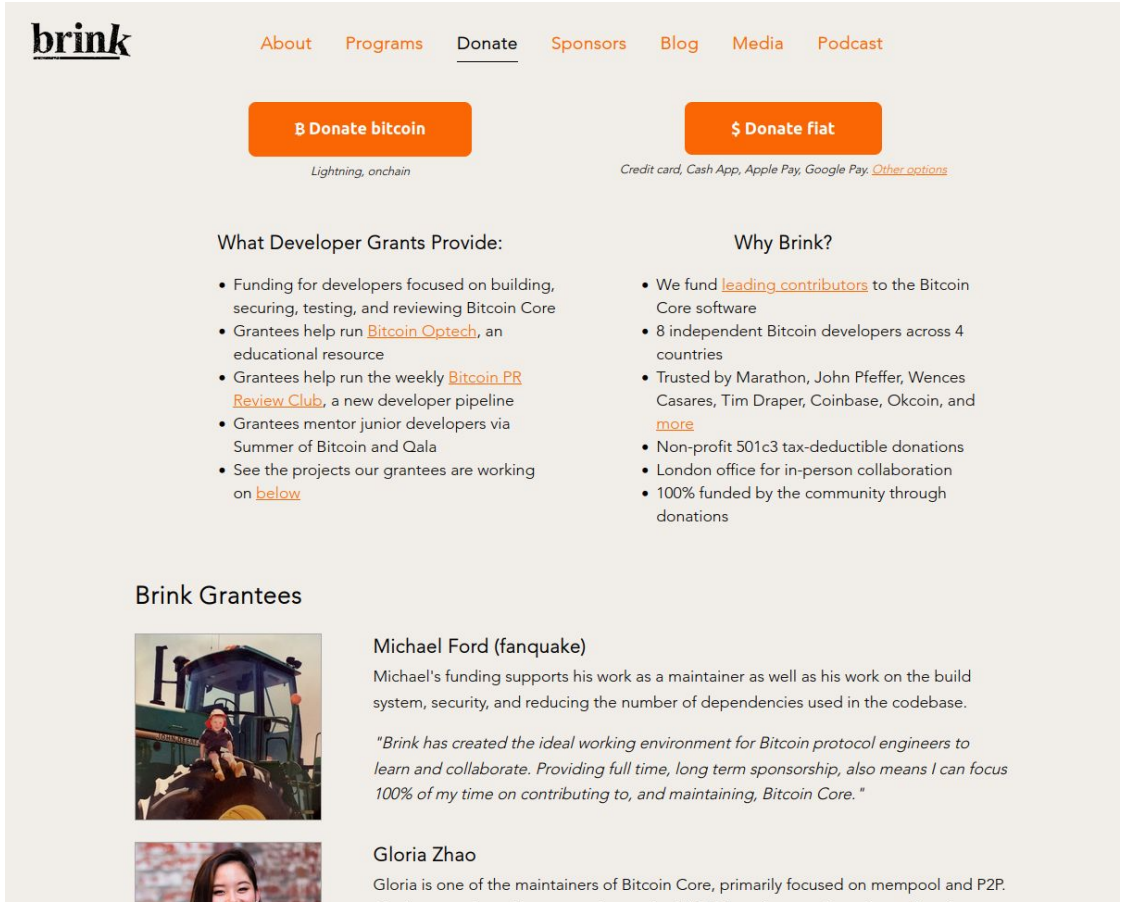
github.com/bitcoin/bitcoin

new devs = **YOU**!



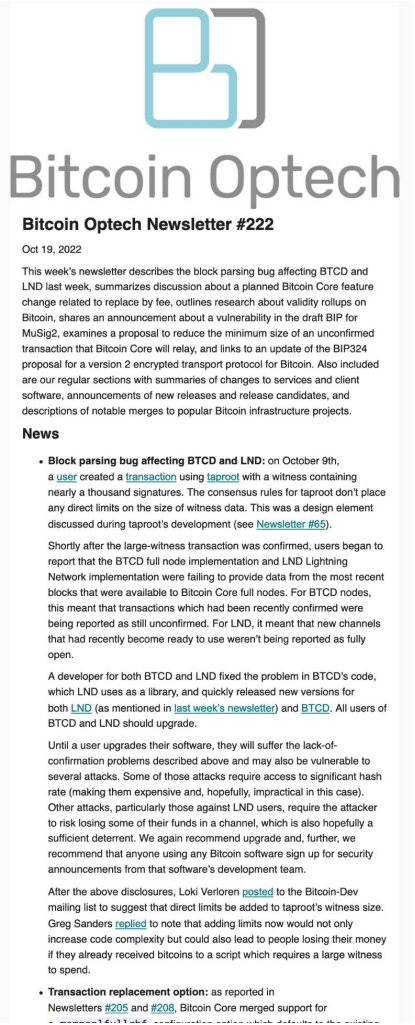
brink.dev/donate

- runs on donations from community
- focus on Bitcoin Core security
- technical grant committee



bitcoinops.org

- Weekly newsletters
- Summaries of topics
- Written/reviewed by devs
- Translated



Bitcoin Optech

Bitcoin Optech Newsletter #222

Oct 19, 2022

This week's newsletter describes the block parsing bug affecting BTCD and LND last week, summarizes discussion about a planned Bitcoin Core feature change related to replace by fee, outlines research about validity rollups on Bitcoin, shares an announcement about a vulnerability in the draft BIP for MuSig2, examines a proposal to reduce the minimum size of an unconfirmed transaction that Bitcoin Core will relay, and links to an update of the BIP324 proposal for a version 2 encrypted transport protocol for Bitcoin. Also included are our regular sections with summaries of changes to services and client software, announcements of new releases and release candidates, and descriptions of notable merges to popular Bitcoin infrastructure projects.

News

- **Block parsing bug affecting BTCD and LND:** on October 9th, a user created a transaction using taproot with a witness containing nearly a thousand signatures. The consensus rules for taproot don't place any direct limits on the size of witness data. This was a design element discussed during taproot's development (see Newsletter #65).

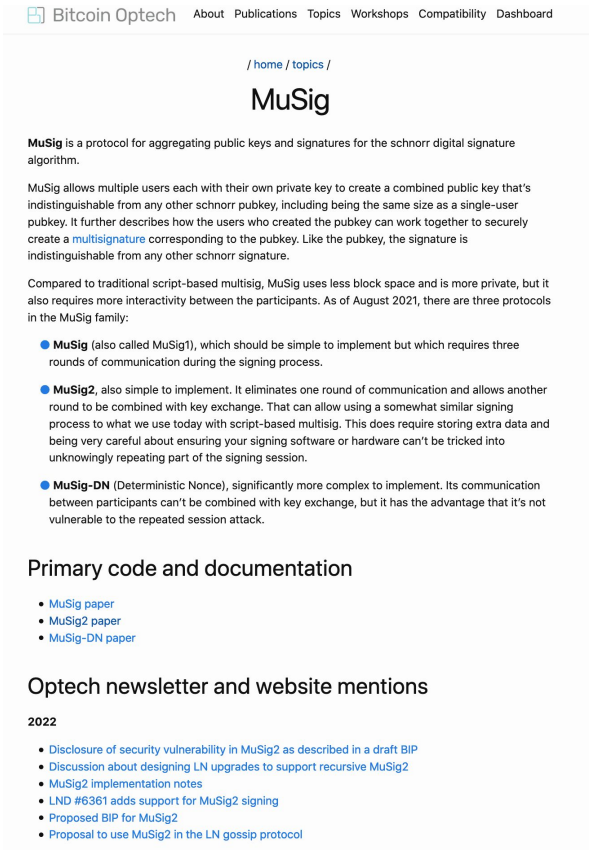
  Shortly after the large-witness transaction was confirmed, users began to report that the BTCD full node implementation and LND Lightning Network implementation were failing to provide data from the most recent blocks that were available to Bitcoin Core full nodes. For BTCD nodes, this meant that transactions which had been recently confirmed were being reported as still unconfirmed. For LND, it meant that new channels that had recently become ready to use weren't being reported as fully open.

  A developer for both BTCD and LND fixed the problem in BTCD's code, which LND uses as a library, and quickly released new versions for both LND (as mentioned in last week's newsletter) and BTCD. All users of BTCD and LND should upgrade.

  Until a user upgrades their software, they will suffer the lack-of-confirmation problems described above and may also be vulnerable to several attacks. Some of those attacks require access to significant hash rate (making them expensive and, hopefully, impractical in this case). Other attacks, particularly those against LND users, require the attacker to risk losing some of their funds in a channel, which is also hopefully a sufficient deterrent. We again recommend upgrade and, further, we recommend that anyone using any Bitcoin software sign up for security announcements from that software's development team.

  After the above disclosures, Loki Verloren posted to the Bitcoin-Dev mailing list to suggest that direct limits be added to taproot's witness size. Greg Sanders replied to note that adding limits now would not only increase code complexity but could also lead to people losing their money if they already received bitcoins to a script which requires a large witness to spend.

- **Transaction replacement option:** as reported in Newsletters #205 and #208, Bitcoin Core merged support for



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MuSig

MuSig is a protocol for aggregating public keys and signatures for the schnorr digital signature algorithm.

MuSig allows multiple users each with their own private key to create a combined public key that's indistinguishable from any other schnorr pubkey, including being the same size as a single-user pubkey. It further describes how the users who created the pubkey can work together to securely create a multisignature corresponding to the pubkey. Like the pubkey, the signature is indistinguishable from any other schnorr signature.

Compared to traditional script-based multisig, MuSig uses less block space and is more private, but it also requires more interactivity between the participants. As of August 2021, there are three protocols in the MuSig family:

- **MuSig** (also called MuSig1), which should be simple to implement but which requires three rounds of communication during the signing process.
- **MuSig2**, also simple to implement. It eliminates one round of communication and allows another round to be combined with key exchange. That can allow using a somewhat similar signing process to what we use today with script-based multisig. This does require storing extra data and being very careful about ensuring your signing software or hardware can't be tricked into unknowingly repeating part of the signing session.
- **MuSig-DN** (Deterministic Nonce), significantly more complex to implement. Its communication between participants can't be combined with key exchange, but it has the advantage that it's not vulnerable to the repeated session attack.

Primary code and documentation

- MuSig paper
- MuSig2 paper
- MuSig-DN paper

Optech newsletter and website mentions

2022

- Disclosure of security vulnerability in MuSig2 as described in a draft BIP
- Discussion about designing LN upgrades to support recursive MuSig2
- MuSig2 implementation notes
- LND #6361 adds support for MuSig2 signing
- Proposed BIP for MuSig2
- Proposal to use MuSig2 in the LN gossip protocol

Thank you!